

Indian Air Quality Monitoring & Pollution Analysis

MongoDB Project Documentation
Environmental Data Analysis using MongoDB

1. Project Overview

This project uses MongoDB to store and analyze daily city-level air-quality observations from India. The analysis focuses on AQI, pollutant levels, city comparisons, AQI categories, yearly trends, and severe pollution observations.

2. Problem Statement

Air-quality datasets contain large numbers of observations across cities and dates. A database system can make these observations easier to query and analyze. This project uses MongoDB aggregation pipelines to identify pollution patterns and compare air quality across cities.

3. Dataset

The project uses the Kaggle Air Quality Data in India (2015–2024) dataset, specifically city_day.csv. After preparation, 18,265 documents were imported into MongoDB.

Field	Purpose
City	Indian city
Datetime	Observation date
PM2_5	PM2.5 concentration
PM10	PM10 concentration
NO, NO2, NOx	Nitrogen-related pollutants
NH3	Ammonia concentration
CO	Carbon monoxide
SO2	Sulfur dioxide
O3	Ozone
Benzene, Toluene, Xylene	Organic pollutants
AQI	Numerical Air Quality Index
AQI_Bucket	Original dataset category
AQI_Category	Validated category derived from AQI

4. Data Quality Check

The original AQI_Bucket field was found to contain inconsistencies with the numerical AQI. It was retained for reference, while a new AQI_Category field was created from the numerical AQI.

AQI Range	Validated Category
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-400	Very Poor

AQI Range	Validated Category
401-500	Severe

5. MongoDB Database

Database: air_quality_db

Collection: city_air_quality

Documents imported: 18,265

6. MongoDB Concepts Demonstrated

- Database and collection creation
- CSV import and document inspection
- countDocuments() and find()
- Projection, sorting, and limiting
- Aggregation pipelines
- \$match, \$group, \$avg, \$sum, \$sort, \$addFields, \$substr, and \$toInt
- Grouping by multiple fields using a nested aggregation key

7. Average AQI by City

City	Average AQI
Mumbai	253.32
Delhi	251.50
Kolkata	250.64
Chennai	250.34
Bangalore	249.77

Result: Mumbai recorded the highest average AQI in the analyzed dataset.

8. Average PM2.5 by City

City	Average PM2.5
Delhi	252.91
Chennai	252.52
Kolkata	250.46
Mumbai	248.56
Bangalore	248.55

Result: Delhi recorded the highest average PM2.5 in the analyzed dataset.

9. AQI Category Distribution

AQI Category	Records
Severe	3,710
Poor	3,645
Very Poor	3,641
Moderate	3,635
Satisfactory	1,822
Good	1,812

Result: Severe was the largest category with 3,710 observations.

10. Average AQI by Year

Year	Average AQI
2015	247.42
2016	250.11
2017	250.89
2018	246.31
2019	255.43
2020	247.31
2021	252.51
2022	250.47
2023	257.01
2024	253.68

The yearly results show variation across the period rather than a consistently improving or worsening pattern.

11. Severe AQI Observations by City

City	Severe Observations
Chennai	766
Mumbai	752
Kolkata	740
Bangalore	738
Delhi	714

Chennai had the highest number of observations classified as Severe using the corrected AQI_Category.

12. Multi-Pollutant Analysis

Average levels were calculated by city for PM2.5, PM10, NO2, SO2, CO, and O3 to provide a broader pollution profile.

13. Project Workflow

- Download and inspect the Kaggle dataset
- Identify the AQI category inconsistency
- Create the validated AQI_Category field
- Import city-level daily observations into MongoDB
- Confirm document count and structure
- Run aggregation queries for city, category, pollutant, and time analysis
- Record results and prepare documentation

14. Key Findings

- Mumbai had the highest average AQI among the analyzed cities.
- Delhi had the highest average PM2.5.
- Chennai had the highest number of Severe AQI observations.
- Severe was the largest corrected AQI category by record count.
- 2023 had the highest annual average AQI in the analyzed period.
- The original AQI_Bucket field contained inconsistencies, making AQI_Category necessary for analysis.

15. Conclusion

The project demonstrates a practical MongoDB workflow for environmental data analysis. MongoDB was used to store 18,265 city-level observations and perform aggregation-based analysis of AQI, pollutant levels, categories, cities, and yearly trends. The project also demonstrates the importance of data-quality validation before using categorical fields for analysis.

16. Project Files

Path	Purpose
data/city_day.csv	Prepared city-level daily air-quality dataset
queries/air_quality_analysis.js	MongoDB analysis queries
documentation/project_documentation.pdf	Project documentation
README.md	GitHub project overview and results

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